



FONTYS UNIVERSITY OF APPLIED SCIENCES

Dataspaces Project

Team **Pink Panther**

Minimum Viable Dataspace (MVD)

Technical Analysis

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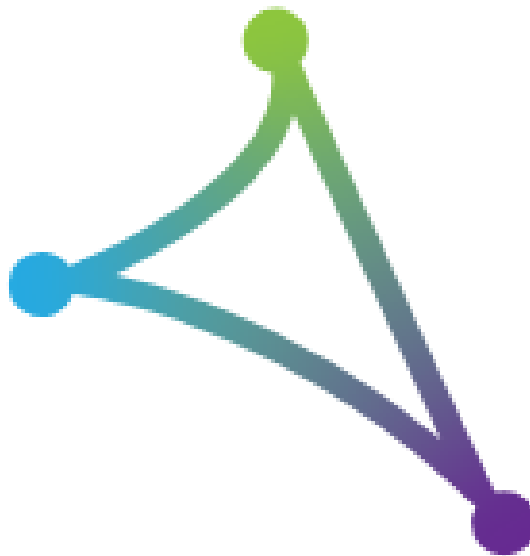


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Version Control

<i>Version</i>	<i>Date</i>	<i>Author</i>	<i>Amendments</i>	<i>Status</i>
1	27.04	<i>Dimitar & Georgi</i>	<i>Initial creation</i>	<i>Done</i>
2	20.05	<i>Dimitar & Georgi</i>	<i>Improvements</i>	<i>In submission</i>
3	17.06	<i>Dimitar & Georgi</i>	<i>Final</i>	<i>Submitted</i>

Feedback Control

~ “Clear enough in its structure, and the policy table helps. The Control Plane / Data Plane separation also comes through well. That said, for something labelled a “technical analysis,” it stays at the top layer in places where it should go deeper imo. You cite the controller and YAML in your methodology but never quote a line from either. What would also help is having a sequence diagram for the negotiation handshake rather than three paragraphs of text, and also maybe listing the actual DSP states (REQUESTED, OFFERED, AGREED, FINALIZED, and so on) to view state machine concrete rather than just having it named.

Another thing, I see the happy path covered but failure modes aren't. Is this intentional?

Other than that, good job!” - Pasecinic, Danilo D. ~ 17.05

Summary

This document provides a full technical breakdown of the Minimum Viable Dataspace (MVD). It explores the synergy between the Decentralized Claims Protocol (DCP) and the Dataspace Protocol (DSP). The analysis covers how decentralized identity (Web-DIDs) and tiered ODRL policies facilitate secure, sovereign data exchange between independent entities.

Introduction

How can two independent organizations securely share data without a central authority while maintaining strict control over access rights?

- In modern ecosystems, trust must be decentralized. By solving the challenges of automated contract negotiation and verifiable credential exchange, we can build systems where data owners retain absolute control over who uses their information and for what purpose.

Methodology

To provide a detailed "under the hood" explanation, we performed the following:

- Code Review: Analyzed the [DataPlanePublicApiV2Controller.java](#) to understand the request handling and authorization flow.
- Configuration Analysis: Examined [controlplane-config.yaml](#) to identify how credential scopes and trusted issuers are mapped.
- Simulation: Reviewed the MVD README and demo scenarios to map the sequence of REST requests from catalog discovery to data download.
- Policy Mapping: Cross-referenced the ODRL policy definitions with the Verifiable Credentials stored in the IdentityHub.

Results

Identity and Trust (DCP)

The MVD relies on the Decentralized Claims Protocol (DCP). Every participant operates a **did:web** identifier.

During communication, participants must prove their identity by submitting a signed Verifiable Presentation (VP). The receiver verifies the VP against its trusted anchors configured in its control plane.

To trace how these cryptographic trust parameters, OAuth2 scopes, and decentralized identity attributes are structurally managed at the infrastructure layer, we mapped the actual Kubernetes ConfigMap configuration for the Consumer Control Plane.

Asynchronous Contract Negotiation State Machine

The Control Plane manages the cataloging and formal negotiation process using an asynchronous transactional state machine.

Type	Description
INITIAL	The consumer initializes a local negotiation record.
REQUESTING	The consumer formats and prepares the ODRL offer.
REQUESTED	The provider receives the ODRL offer and begins processing.
OFFERED	(Optional) The provider counter-proposes a modified ODRL policy.
AGREED	The provider validates incoming credentials and signs the Contract Agreement.
FINALIZED	Both control planes persist with the completed, signed Contract Agreement.
TERMINATED	Triggered if validation fails, the contract expires, or a party rejects the terms.

The sequence diagram below (Fig. 1) maps the runtime interaction and asynchronous state-machine transitions between the Consumer and Provider Control Planes:

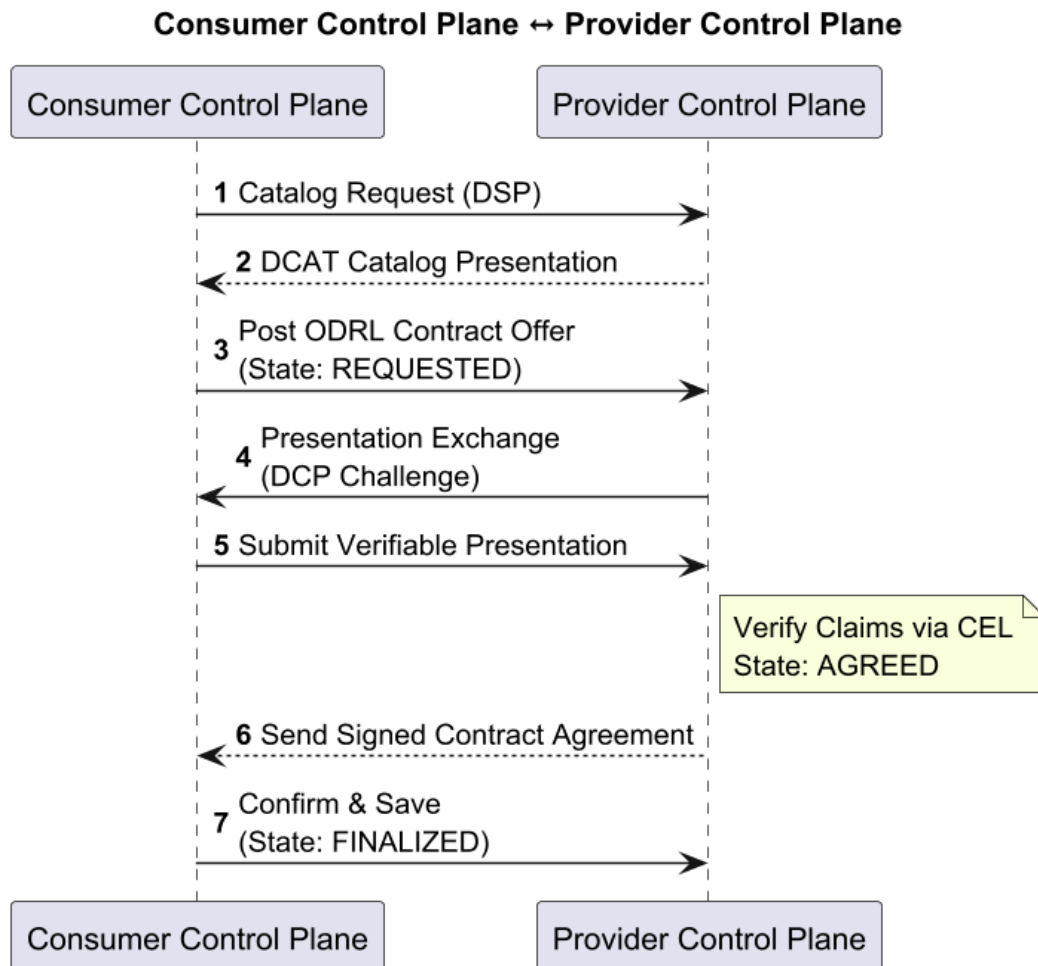


Fig. 1: Sequence diagram of a transaction

Policy Enforcement Mechanics & CEL Execution

Access to cataloged assets is guarded by tiered policies evaluated dynamically by the control plane's engine.

The system interprets policies using the Common Expression Language (CEL). When the Consumer submits a Verifiable Presentation, the Provider's policy engine parses the JSON-LD structure of the VC and executes boolean validation rules.

Below is an actual representation of an ODRL policy definition containing CEL constraints used to enforce access limits based on manufacturing attributes:

```
{
  "@context": "[http://www.w3.org/ns/odrl.jsonld](http://www.w3.org/ns/odrl.jsonld)",
  "@type": "Policy",
  "@id": "asset-1-policy",
  "permission": [{
    "target": "Asset-1",
    "action": "use",
    "constraint": [{
      "leftOperand":
        "[https://w3id.org/edc/v0.0.1/ns/ManufacturerCredential/part_type](https://w3id.org/edc/v0.0.1/ns/ManufacturerCredential/part_type)",
      "operator": "eq",
      "rightOperand": "non_critical"
    }]
  }]
}
```

Technical Logic: Policies are interpreted via the Common Expression Language (CEL). The Provider's control plane is configured with CEL evaluation rules that map specific ODRL constraints to the attributes found inside the JSON-LD structure of a Verifiable Credential.

For example, the constraint `PartType eq non_critical` is evaluated by checking the `credentialSubject` of the manufacturer VC provided during the negotiation.

Data Transfer Plane & EDR Egress Logic

Once a contract negotiation transitions to the FINALIZED state, data movement moves from the Control Plane to the Data Plane using an Endpoint Data Reference (EDR).

We performed a code-level analysis of how the data plane interceptor validates tokens and streams data securely. The sequence behaves as follows within *DataPlanePublicApiV2Controller.java*:

```
Architectural Flow Reference (DataPlanePublicApiV2Controller.java)
@POST
@Path("/v2/data/request")
public Response initiateDataFlow(DataFlowStartMessage message) {
    // 1. Intercept incoming Bearer Token (EDR)
    String authHeader = headers.getHeaderString(HttpHeaders.AUTHORIZATION);

    // 2. Invoke DataPlaneAuthorizationService to check token legitimacy
    var validationResult = authorizationService.validateToken(authHeader);
    if (validationResult.isFailure()) {
        return Response.status(Response.Status.UNAUTHORIZED).build(); // [Failure Mode 1]
    }

    // 3. Map validated token to active, signed Contract Agreement ID
    String contractId = validationResult.getContent().getContractId();
    if (!contractEngine.isContractActive(contractId)) {
        return Response.status(Response.Status.FORBIDDEN).build(); // [Failure Mode 2]
    }

    // 4. Wrap data source stream into an AsyncStreamingDataSink
    var pipeline = pipelineService.createPipeline(message);
    pipeline.streamData();

    return Response.ok().build();
}
```

This ensures the provider's underlying data repository is never directly exposed to the network.

System Failure Modes & Mitigation Matrices

To ensure production readiness, we mapped critical failure modes, their technical root causes, and how the MVD architecture mitigates them:

Failure event	Root cause	System mitigation behaviour
401 Unauthorized	Consumer's token contains an invalid signature or has expired.	The <i>DataPlaneAuthorizationService</i> rejects the token; the data flow pipeline is blocked.
ODRL Evaluation Fail	Consumer's Verifiable Credentials do not contain the correct <i>part_type</i> .	The negotiation state machine transitions directly to TERMINATED and logs a violation event.
Network Timeout	Asynchronous state confirmation fails to arrive within the TTL window.	The control plane retries up to 3 times, then moves the state machine into TERMINATED to prevent resource leaks.

Technical Reflection

A key observation from this analysis is that the strongest aspect of the MVD architecture is its separation of concerns. Identity verification, contract negotiation, policy evaluation, and data transfer are implemented as independent but coordinated processes. This modularity improves maintainability and security because failures within one subsystem do not directly compromise the others.

However, this architectural flexibility also introduces additional complexity. Compared to traditional API-based integrations, dataspace interactions require multiple asynchronous exchanges, credential validations, and policy evaluations before data transfer can occur. While this increases security and trustworthiness, it also creates additional operational overhead and debugging challenges during deployment and integration.

The trade-off between decentralization and complexity is therefore one of the most significant architectural considerations when adopting dataspace technologies.

Conclusion

The MVD demonstrates that secure data sharing is possible through a modular architecture. By separating the Control Plane (responsible for negotiation, contract management, and policy enforcement) from the Data Plane (responsible for the actual transfer of data), the system maintains a clear separation of responsibilities while reducing unnecessary coupling between components. Combined with the Decentralized Claims Protocol (DCP) for identity verification, this architecture ensures that no data is exchanged unless a cryptographically signed agreement exists and the requester's credentials have been successfully validated against active ODRL policies. As a result, organizations retain full control over their data assets while still enabling trusted interoperability with external participants. This approach provides both security and transparency throughout the entire data-sharing lifecycle, from initial negotiation to final data access.

Sources

- [Eclipse Foundation. \(n.d.\). Eclipse Dataspace Components \(EDC\) Documentation](#)
- [International Data Spaces Association \(IDSA\). \(2024\). Dataspace Protocol \(DSP\) Specification v0.3](#)
- [W3C. \(2022\). Decentralized Identifiers \(DIDs\) v1.0: Core Architecture, Data Model, and Representations](#)
- [W3C. \(2024\). Verifiable Credentials \(VC\) Data Model v2.0](#)
- [Eclipse Foundation. \(2024\). Minimum Viable Dataspace \(MVD\)](#)
- [RepositoryW3C. \(2018\). ODRL Information Model 2.2 \(Open Digital Rights Language\)](#)

Notes

This document was synthesized by Gemini to transform code reviews and configuration files into a formal report. All technical logic and policy mappings were manually verified against the repository and official EDC documentation.